

1. **Simplify** the following boolean expression: $(A + B)(C+B)(C+A)$

2. **Find** the complement of the following expression:

$$(x + y' + z')(x' + y)(x' + z')$$

3. **Draw** the following functions using NOR gates only:

$$F(A,B,C,D) = (A'BCD' + A'C' + A(B'+D'))$$

NB: You can't simplify the above function. You have to draw based on the function given in question.

4. **(Ungraded) Draw** the following functions using NAND gates only:

$$F(A,B,C,D) = (A' B' + C'D + CD + (B'C + D'))$$

NB: You can't simplify the above function. You have to draw based on the function given in question.

Find out SOP and POS for the following: (Do not use truth table)

$$F(A,C,B) = A'C + BC'$$

6. $F(A,B,C,D) = A + B'C' + D$

7. (Ungraded) $F(A,B,C,D,E) = ACE + CDE$